CLASS : T. E. E&TC		**SUBJECT:** CN LAB
EXPT. NO. : 10	**Roll No.:**	**DATE :**

TITLE:

Effects of propagation and B.S. configuration on C/I distribution

OBJECTIVES:

1. Cell radios,
2. Tx power of B.S,
3. Frequency reuse,
4. Sectoring,
5. Shadowing effect,
6. B.S. height,
7. Path loss exponent,
8. Vertical beam tilt

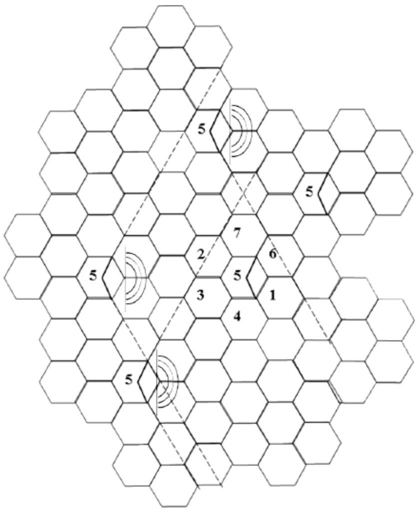
S/W USED:

VLAB

THEORY:

Concepts of path loss, shadowing, antenna height, horizontal beam pattern, vertical beam tilt boundary coverage probability and calculation of SINR have been explained in details. Along with these, the concept of clustering and frequency reuse has also been explained. This particular experiment combines all the previous concepts which together comprise a basic cellular mobile communication system. Therefore it is a pre-requisite that all previous experiments be completed before starting on this.

One of the concepts introduced here is sectoring.



How 120 degree sectoring reduces interference from co-channel cells. Out of the 6 co-channel cells in the first tier, only two of them interfere with the center cell. If omnidirectional antennas were used at each base station , all sixco-channel cells would interfere with the center cell.

It can be easily seen that the number of interfering sites drastically reduce for a given reuse factor. In effect, the carrier to interference ratio at a cell edge in downlink becomes.

$$\frac{P_T \cdot d^{-n_p}}{P_T \sum_{i=1}^{2} d_i}$$

$$\frac{P_T \cdot d^{-n_p}}{P_T \sum_{i=1}^{6} d_i}$$

whereas for Omni directional cell the C/I is

$$\frac{P_T \cdot d^{-n_p}}{P_T \sum_{i=1}^{2} d_i}$$

$$\frac{P_T \cdot d^{-n_p}}{P_T \sum_{i=1}^{6} d_i}$$

Thus there in a 3 folds increase in C/I by virtue of sectoring.

Thus SINR at cell edge can be improved. Since a service does not require an SINR greater than a certain threshold, therefore by using sectoring the cluster size can be reduced thereby increasing capacity.

Now in this experiment since all possible parameters have been considered, it is a complex mix of several effects. Therefore the student is urged to vary different parameters and compare the effects on distribution of C/I.

ADVANTAGES/ DISADVANTAGES (IF ANY):

ADVANTAGES:

1. Improved Carrier-to-Interference Ratio (C/I)
2. Increased System Capacity
3. Better Coverage Control
4. Flexibility in Network Design
5. Reduced Interference with Sectoring
6. Improved Quality of Service (QoS)

DISADVATAGES:

1. Increased System Complexity
2. Higher Cost
3. Inter-Sector Interference
4. Sensitivity to Environmental Factors
5. Optimization Challenges
6. Handover Complexity

SAMPLE CALCULATIONS /OBSERVATIONS (USING CALCULATOR):

- **CONCLUSION:**

Propagation conditions and base station configuration have a major impact on the C/I distribution in a cellular system. By adjusting parameters like transmit power, antenna height, frequency reuse, and especially sectoring, interference can be reduced and signal quality improved. Sectoring significantly enhances C/I, allowing higher capacity and better service quality.

However, these improvements increase system complexity and cost, and performance can still be affected by shadowing and improper parameter settings.

In conclusion, proper configuration of base station parameters improves C/I and system capacity, but requires a careful balance between performance, cost, and complexity.

FACULTY SIGN:	**REMARK:**